

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Engineered **Python** data pipelines to ingest and transform ~500GB of daily advertising performance data, creating feature sets for ML model training and reducing data latency by **40%**.
- Developed and deployed **Java Spring Boot** microservices for real-time bidding and ad-serving logic, handling over **2,000 requests per second** during peak loads.
- Designed and implemented **ML algorithms** to optimize ad spend ROI, focusing on predictive modeling for audience segmentation and campaign performance forecasting.
- Collaborated with Data Engineers to integrate novel ML algorithms into production pipelines on **AWS**, ensuring scalability and reliability for millions of daily decisions.
- Wrote production code for **feature engineering**, transforming raw traffic and user interaction data into meaningful inputs for machine learning models, improving model accuracy by **15%**.
- Trained and evaluated **gradient-boosted models** (XGBoost) on a dataset of **1.2M user interactions**, achieving an AUC of **0.88** for predicting ad click-through rates.

SKILLS

Machine Learning – ML algorithms, predictive modeling, feature engineering, model training, XGBoost, scikit-learn, statistics, optimization

Programming Languages – Python, Java, SQL, R

Backend Development – Spring Boot, REST APIs, microservices, FastAPI, Flask

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, CI/CD, Jenkins, Git

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, data modelling, data warehousing, NumPy, pandas

Databases – PostgreSQL, MySQL, Oracle

Systems & Operations – Linux, Splunk, Dynatrace, incident response, change management

PROJECTS

Real-Time Bidding ML Model Optimization

- Developed a **Python-based feature pipeline** to extract user behavioral signals and contextual data from millions of ad impression events.
- Trained an **XGBoost classification model** to predict bid success probability, achieving **0.91 AUC** on a held-out test set representing 3 days of traffic.
- Packaged the trained model into a **FastAPI service** with input validation and basic monitoring, integrating it into an existing bidding system for A/B testing.

Audience Segmentation for Targeted Advertising

- Applied **unsupervised learning techniques (K-Means clustering)** in R to segment a user base of **150,000 users** based on browsing history and campaign engagement.
- Generated **Power BI dashboards** visualizing segment characteristics and campaign performance per segment, aiding marketing analysts in campaign strategy.
- Collaborated with marketing stakeholders to refine segmentation criteria based on feedback and observed campaign uplift of **~12%** in pilot campaigns.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)